25901435 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCPHY200 - WAVE OPTICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. What is the relation between wave vector and wavelength? [U] / [CO1]
2. Formulate the wave equation in three dimensions. [K] / [CO1]
3. Mention any four quantities that describe a wave. [K] / [CO1]
4. In a polarising device, what is referred to as the extinction axis? [U] / [CO2]
5. Why is quartz said to be optically active? [U] / [CO2]
6. What phase difference corresponds to a path difference of λ/2? [U] / [CO3]
7. Explain the formation of Newton’s rings. [K] / [CO3]
8. Write down the mathematical expression for the resultant irradiance in a double slit experiment and define the symbols used. [U] / [CO3]
9. Why do two separate light sources not normally generate sustained interference fringes? [U] / [CO3]
10. Why is precise alignment important in the Michelson interferometer? [A] / [CO3]

11. What are the conditions for secondary maxima in a multi-slit Fraunhofer diffraction pattern? [U] / [CO4]

12. What is the path difference between two consecutive half-period zones? [U] / [CO4]

13. What are the necessary conditions for obtaining Fraunhofer diffraction? [U] / [CO4]

14. Discuss the dependence of the width of the central diffraction maximum on the wavelength of light in single-slit diffraction. [A] / [CO4]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Using Huygens' construction, explain the formation of ordinary and extraordinary wavefronts in a uniaxial crystal. [U] / [CO2]

16. Explain polarisation by scattering. [K] / [CO2]

17. Explain phasor representation of harmonic waves. Show how two sinusoidal waves of the same frequency but different phases can be added using a phasor diagram. [U] / [CO3]

18. A wire is placed between two glass plates forming an air wedge. Derive an expression for calculating its thickness. [U] / [CO3]

19. Discuss the interference of reflected light from the thin film, Deduce the condition for constructive and destructive interference. [A] / [CO3]

20. Explain the Huygens - Fresnel principle and describe how it accounts for the propagation of light waves. [U] / [CO4]

21. Sketch and compare the fringe patterns obtained in Young's double-slit experiment and in Fraunhofer double-slit diffraction [A] / [CO4]

22. Describe the construction of a zone plate. [U] / [CO4]

[5x6 = 30]